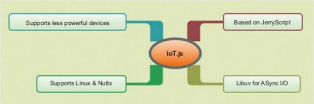

Figure 4: IoT.js

If you want to explore Favor in detail, then visit the official Favor wiki at *https://github.com/favor/it/wiki*.

JerryScript

JerryScript is a lightweight engine for JavaScript. Its prime advantage is its ability to run in resource-constrained devices. JerryScript can run on devices with less than 64KB of RAM. It even supports devices with less than 200KB of Flash memory (*https://github.com/jerryscript-project/jerryscript*).

The major highlights of JerryScript are listed below (Figure 3):

- Optimised for microcontrollers.
- Code is actively updated at frequent intervals.
- JerryScript is ECMAScript 5.1 standards compliant.
- Ability to precompile the JavaScript source code to byte code.
- Highly optimised for low memory consumption.
- JerryScript is open source and available with the Apache License 2.0.
- Written in C99 for maximum portability.

JerryScript was developed from scratch by the Samsung Open Source Group.

Detailed instructions regarding pre-requisites and setting up are provided at *https://github.com/jerryscript-project/jerryscript/blob/master/docs/01.GETTING-STARTED.md*.

IoT.js

IoT.js is also targeted towards resource-poor devices; it is built on top of JerryScript. Its features are:

- The IoT.js platform uses JerryScript to execute JavaScript.
- It uses *libuv* for asynchronous input and output (*https://github.com/Samsung/libuv*). *libuv* is a multi-platform support library to enable asynchronous I/O.
- Presently, IoT.js supports Linux and NuttX. The latter is a real-time operating system.
- The API provides features to handle buffers, events, consoles, GPIO, streams, timers, etc.

Detailed information regarding app development using IoT.js is available at *https://github.com/Samsung/iotjs*.

Cyclon

Cyclon.js is a JavaScript framework to support robotics, physical computing and IoT. One of its major advantages is the support for various platforms. As per the official documentation (*https://cylonjs.com/*) it supports 43 different platforms.

Getting started with Cyclon.js is straightforward. Just install the *npm* module, as follows:

```
npm install cyclon
```

A sample code snippet to toggle an LED every three seconds is shown below:

```
var Cylon = require("cylon");

// Initialize the robot

Cylon.robot({

// Change the port to the correct port for your Arduino.
connections: {
arduino: { adaptor: 'firmata', port: '/dev/ttyACM0' }
},

devices: {
   led: { driver: 'led', pin: 13 }
},

work: function(my) {
        every((3).second(), function() {
               my.led.toggle();
        });
}
}).start();
```

To run the code, type:

```
$ npm install cylon-firmata cylon-gpio cylon-i2c
$ node script.js
```

Cyclon.js can be executed from a browser with the help of the *browserify* module of *npm*. END

References

[1] *https://www.postscapes.com/javascript-and-the-internet-of-things/*
[2] *https://github.com/favor/it*
[3] *http://iotjs.net/*
[4] *https://github.com/Samsung/iotjs*
[5] *https://github.com/jerryscript-project/jerryscript*
[6] *https://cylonjs.com/*
[7] 'JavaScript on Things', a book by L.D. Gardner

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